

MODULE 01:

1. Designer Door Mat: (full code)

```
n, m = map(int, input().split())
pattern = [('.')*(2*i + 1)).center(m, '-') for i in range(n//2)]
print('\n'.join(pattern + ['WELCOME'.center(m, '-')] + pattern[::-1]))
```

2. Find a String:

```
def count_substring(string, sub_string):
    count = 0
    for i in range(len(string)):
        if string[i:].startswith(sub_string):
            count += 1
    return count

if __name__ == '__main__':
    string = input().strip()
    sub_string = input().strip()

    count = count_substring(string, sub_string)
    print(count)
```

3. Swap Case:

```
def swap_case(s):
    res = ""
    for c in s:
        if c.isalpha and c.islower():
            res += c.upper()
        elif c.isalpha and c.upper():
            res += c.lower()
        else:
            res += c
    return res

if __name__ == '__main__':
    s = input()
    result = swap_case(s)
    print(result)
```

4. Nested Lists: (full code)

```
if __name__ == '__main__':
    # Step 1: Take input
    students = []
    for _ in range(int(input("Enter number of students: "))):
        name = input("Enter name: ")
        score = float(input("Enter score: "))
        students.append([name, score])
```

```

# Step 2: Extract all scores
scores = [s for name, s in students]

# Step 3: Find the second lowest score
second_lowest = sorted(set(scores))[1]

# Step 4: Print names with that score (sorted alphabetically)
for name, score in sorted(students):
    if score == second_lowest:
        print(name)

```

5. List Comprehensions:

```

if __name__ == '__main__':
    x = int(input())
    y = int(input())
    z = int(input())
    n = int(input())

    print([i,j,k for i in range(x+1) for j in range(y+1) for k in
    range(z+1) if(i+j+k) != n ])

```

MODULE 02:

1. Alphabet Rangoli:

```

import string
def print_rangoli(n):

    alpha = string.ascii_lowercase

    L = []
    for i in range(n):
        s = "-".join(alpha[i:n])
        L.append((s[::-1]+s[1:]).center(4*n-3, "-"))
    print('\n'.join(L[:0:-1]+L))

if __name__ == '__main__':
    n = int(input())
    print_rangoli(n)

```

2. String Validators:

```

if __name__ == '__main__':
    s = input()

    print(any(c.isalnum() for c in s))
    print(any(c.isalpha() for c in s))
    print(any(c.isdigit() for c in s))
    print(any(c.islower() for c in s))
    print(any(c.isupper() for c in s))

```

3. The Captain's Room: (full code)

```
k = int(input())
lst = list(map(int, input().split()))
captain = (sum(set(lst)) * k - sum(lst)) // (k - 1)
print(captain)
```

4. Time Delta

```
import math
import os
import random
import re
import sys
from datetime import datetime

def time_delta(t1, t2):
    fmt = "%a %d %b %Y %H:%M:%S %z"
    dt1 = datetime.strptime(t1, fmt)
    dt2 = datetime.strptime(t2, fmt)
    k = (abs(dt1-dt2).total_seconds())
    return str(int(k))

if __name__ == '__main__':
    fptr = open(os.environ['OUTPUT_PATH'], 'w')
    t = int(input())
    for t_itr in range(t):
        t1 = input()
        t2 = input()
        delta = time_delta(t1, t2)
        fptr.write(delta + '\n')
    fptr.close()
```

5. Map and lambda function:

```
cube = lambda x:x ** 3
def fibonacci(n):
    # return a list of fibonacci numbers
    fib_list=[]
    a,b =0,1
    for _ in range(n):
        fib_list.append(a)
        a,b=b,a+b
    return fib_list
if __name__ == '__main__':
    n = int(input())
    print(list(map(cube, fibonacci(n))))
```

Module - 03

1. Map and lambda Function (same as the 5th program of M2)

2. Validating CC Numbers: (full code)

```

import re

TESTER = re.compile(
    r"^"                # start of string
    r"(?!.*(\d)(-?\1){3})" # no four same digits in a row (e.g. 1-1-1-1)
    r"[456]"            # must start with 4, 5, or 6
    r"\d{3}"             # next 3 digits (first 4 total)
    r"(?:-?\d{4}){3}"    # three groups of 4 digits
    r"$"                # end of string
)

for _ in range(int(input().strip())):
    print("Valid" if TESTER.search(input().strip()) else "Invalid")

```

3. No Idea :(full code)

```

n, m = map(int, input().split())
lst = list(map(int, input().split()))
A, B = set(map(int, input().split())), set(map(int, input().split()))
print(sum((i in A) - (i in B) for i in lst))

```

4. Word Order :(full code)

```

from collections import Counter, OrderedDict
class OrderedCounter(Counter, OrderedDict):
    pass
d = OrderedCounter(input() for _ in range(int(input())))
print(len(d))
print(*d.values())

```

5. Find the runner up score:

```

if __name__ == '__main__':
    n = int(input())
    arr = map(int, input().split())
    arr = set(arr)
    arr.remove(max(arr))
    print(max(arr))

```
